

# Characterization of fast steering mirrors for free space communication

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## ABSTRACT

The low orbit of Earth is currently experiencing a surge in the number of small cube-satellites, the majority of which are utilised for communication purposes. Initially, these satellites were linked via radio frequency (RF) connections, but in recent times, optical links utilising lasers have become the norm due to their superior data transmission rates. But how can the laser on one cube-sat be consistently directed towards another, while both satellites orbit around earth? This is made possible by a fast steering mirror, which is able to redirect the laser beam with high frequency and accuracy. It enables the correction of vibrations and of micro-movements of the satellite as well as adjustments that consider the time of flight of the laser light.

The distance between the sending and receiving cube-sats is typically several thousand kilometres. Any deformations of the fast steering mirror's surface shape will directly imprint on the wavefront of the reflected laser beam, which in turn will increase the laser beam diameter due to propagation along the entire path from sending to receiving cube-sats. This will reduce the quality of the data transmission.

This paper presents a discussion and comparison of two different measurement approaches for the qualification of the surface flatness of the fast steering mirror, deflectometry and wavefront sensor respectively. These two technologies are complementary in this application and both offer high accuracy at  $\lambda/10$  pTV or better for flat mirror testing. While deflectometry generally offers a higher lateral resolution, wavefront sensors measure at a higher speed and in single shot.

The analysis encompasses measurements of the influence of the fast movement and corresponding acceleration on the fast steering mirror's surface shape aberrations. Furthermore, we discuss the behaviour under normal ambient conditions as well as in vacuum conditions.

**Keywords:** Free space communication, Fast steering mirrors, Wavefront testing, Surface shape measurement, Phase shifting deflectometry

## 1. INTRODUCTION

The number of Cube-sats present in low earth orbit is increasing constantly and costs for the launch of small satellites are decreasing [1,2]. The tasks assigned to cube-sats are manifold, however, free space communication is surely one of the most common and primary important one these days. Advances in laser technology and data processing enable communication between cube-sats in free space over large distances of several thousands of kilometers [3]. To achieve this, the laser beam must be directed with utmost precision and stability. The photon detector at the receiving satellite has typical dimensions of 10mm in diameter, so laser beam pointing angle errors of a few micro rad already lead to a loss of the signal and breakdown of the communication between the satellites. Therefore, vibrations and micro-movements of the cube-sat have to be minimized. This is achieved by a fast steering mirror (FSM) [4]. In addition to the high steering precision and stability, an optimal surface quality of the mirror is required: Deformations of the FSM's surface shape will directly imprint on the wavefront of the reflected laser beam. When the wavefront propagates from the sending to the receiving cube-sat, such wavefront aberrations are continuously transferred to the laser beam intensity [5] and will thus deform and increase the laser beam diameter along the beam path. Consequently, a beam with reduced intensity and inhomogeneous intensity distribution arrives at the receiver, reducing the quality of the data transmission. A constant high quality of the FSM can only be achieved by rigorous quality control, i.e. dedicated measurement of the FSM surface shape. In section 2, two methods for the assessment of the mirror surface shape aberrations are discussed and compared: phase shifting deflectometry [6] and reflected wavefront testing using Shack-Hartmann wavefront metrology [7]. Section 3 provides an analysis of the FSM dynamic behaviour of the FSM and shows measurement results on the surface shape

deformation acquired during actuation of the FSM. Also, the surface shape measurement under vacuum conditions is discussed.

## 2. FSM SURFACE SHAPE CHARACTERIZATION

One of the key components of the FSM is the flat mirror. For successful application in free space communication, a high surface flatness with surface shape deviations of few nm RMS it is of utmost importance in order to keep the reflected laser beam as collimated as possible. The measurement of the surface shape needs to be performed in a contact-free manner, since we need a non-destructive testing. In this work, we applied phase shifting deflectometry and Shack-Hartmann wavefront metrology to assess the surface quality of the FSM. Both methods can provide the surface shape of optical surfaces with high precision and accuracy.

### Phase shifting deflectometry

As shown in figure 1, a high-quality screen projects high-contrast sine-wave patterns onto the mirror under test. The mirror reflects the pattern, thereby changing the features contained in the pattern according to the mirror's surface shape. Two stereo cameras capture the reflected pattern and a 3D point cloud of the mirror surface is generated based on the two images parallelly acquired. That way, the surface shape of the FSM is acquired with a lateral spatial resolution of  $100\mu\text{m}$ . The time for image acquisition is ca. 2sec, then the data is processed within typ. 3sec.

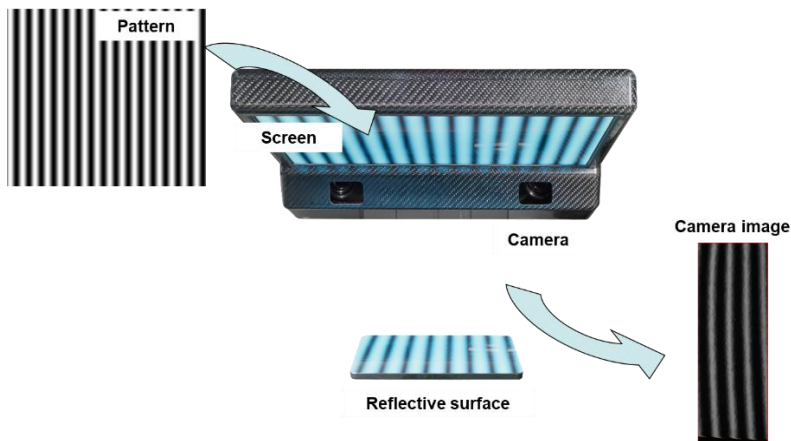


Figure 1: Setup for testing of reflective surfaces by using phase shifting deflectometry done with a reflectCONTROL 130-160 3D HLP.

### Shack-Hartmann wavefront metrology

The light emitted by a fibre coupled LED with central wavelength 530nm is collimated and directed towards the FSM under test as shown in Figure 2. A Keplerian telescope images the light reflected from the FSM to the Shack-Hartmann wavefront sensor. The FSM and the micro lens array are located in the conjugate planes of the telescope, in order to avoid loss of accuracy due to wavefront propagation.

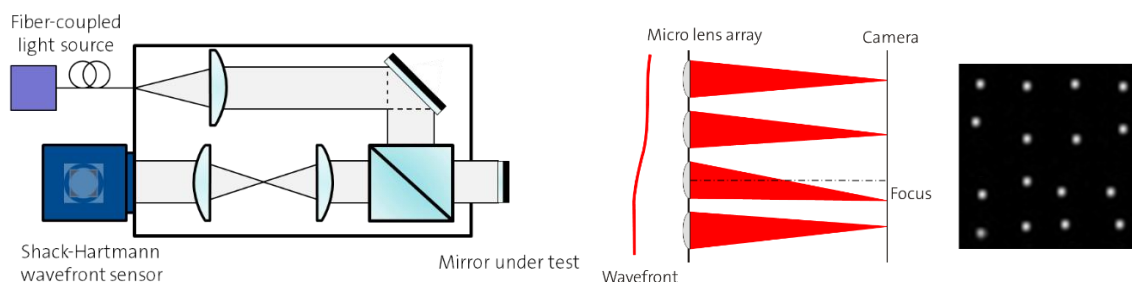


Figure 2: Setup for testing of plano mirrors in reflected light configuration using a Shack-Hartmann wavefront sensor

The Shack-Hartmann wavefront sensor uses a micro lens array with 203 x 203 micro lenses and a micro lens pitch of 75 $\mu$ m. The spot displacement at one micro lens (lateral distance of the focus point from the optical axis of the micro lens) corresponds to the local wavefront gradient. By numerical integration of the field of spot displacements, the 2D wavefront is calculated. Then a Zernike polynomial is fit to the reconstructed wavefront, providing detailed information on the type of aberrations contained in the reflected wavefront. In the measurement configuration as shown above in Figure 2, the reflected wavefront directly equals the surface shape deviation, we only need to apply a factor of 2 introduced by the reflection.

### Surface shape measurement results and comparison

In this study we measured three different mirrors with different magnitude of surface shape aberrations shown in Figure 3 using the above described measurement methods.



Figure 3: Mirrors measured in this study: 1- Gold coated mirror, 2 – Zerodur mirror with aluminum coating, 3 - molybdenum mirror

The 2D surface aberration maps resulting from the measurements of the gold coated mirror using the wavefront sensor (WFS) and the phase shifting deflectometer (PSD) are shown in **Fehler! Verweisquelle konnte nicht gefunden werden..** Both measurement approaches show a qualitatively similar surface shape of the mirror. It is evident how the surface aberrations rotate along when the mirror is rotated in steps of 90°. Slight differences between the measurements obtained from the PSD and the WFS show in the peak-to-valley (PV) and root mean square (RMS) values of the surface shape aberration maps.

Table 1: 2D surface aberration maps obtained from measurement of the gold coated mirror, measured in different rotational orientations.

|     | 0°                      | 90°                     | 180°                    | 360°                    |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| PSD |                         |                         |                         |                         |
|     | PV: 60nm<br>RMS: 10.0nm | PV: 72nm<br>RMS: 12.3nm | PV: 57nm<br>RMS: 9.1nm  | PV: 63nm<br>RMS: 10.0nm |
| WFS |                         |                         |                         |                         |
|     | PV: 72nm<br>RMS: 10.7nm | PV: 73nm<br>RMS: 10.9nm | PV: 74nm<br>RMS: 11.0nm | PV: 70nm<br>RMS: 10.6nm |

Measurements have been carried out under several rotational orientations, also with the molybdenum mirror and the Zerodur mirror. Table 2 summarizes the surface data obtained from the measurement of the three mirrors. The differences between the PSD and the WFS measurement approach are in the range of a few nm, which is close to the repeatability of each method for itself. Hence, we can state that both approaches show a good general agreement on the measured 2D surface shape maps of the three mirrors tested in this study.

Table 2: Comparison of the PV and RMS values of the surface shape measurements obtained with the PSD and WFS measurement approaches for the three mirrors. Each mirror was measured in four rotational orientations.

| Gold           | PSD       |            | WFS       |             | Zerodur        | PSD       |            | WFS       |            | Molybdenum     | PSD        |              | WFS        |            |
|----------------|-----------|------------|-----------|-------------|----------------|-----------|------------|-----------|------------|----------------|------------|--------------|------------|------------|
|                | PV [nm]   | RMS [nm]   | PV [nm]   | RMS [nm]    |                | PV [nm]   | RMS [nm]   | PV [nm]   | RMS [nm]   |                | PV [nm]    | RMS [nm]     | PV [nm]    | RMS [nm]   |
| <b>0°</b>      | 60        | 10         | 72        | 10.7        | <b>0°</b>      | 25        | 0.9        | 18        | 2.6        | <b>0°</b>      | 475        | 119.6        | 461        | 121.1      |
| <b>90°</b>     | 72        | 12.3       | 73        | 10.9        | <b>90°</b>     | 23        | 2.7        | 19        | 2.8        | <b>90°</b>     | 472        | 119.7        | 462        | 120.4      |
| <b>180°</b>    | 57        | 9.1        | 74        | 11          | <b>180°</b>    | 27        | 4          | 17        | 2.7        | <b>180°</b>    | 463        | 119.2        | 474        | 121.6      |
| <b>360°</b>    | 63        | 10         | 70        | 10.6        | <b>360°</b>    | 26        | 3.2        | 19        | 3.1        | <b>360°</b>    | 470        | 119.5        | 460        | 120.5      |
| <b>Average</b> | <b>61</b> | <b>9.9</b> | <b>72</b> | <b>10.8</b> | <b>Average</b> | <b>24</b> | <b>2.8</b> | <b>19</b> | <b>2.8</b> | <b>Average</b> | <b>472</b> | <b>119.7</b> | <b>461</b> | <b>120</b> |

### 3. DYNAMIC MEASUREMENT OF THE FSM SURFACE SHAPE ABERRATIONS

The wavefront sensor reconstructs the reflected wavefront from one single camera image. This allows the analysis of the FSM surface shape aberrations during actuation of the FSM. The FSM can drive the mirror with oscillation frequencies up to 5kHz. At low frequencies, the beam deflection angle can reach values larger than  $\pm 1^\circ$ . With increasing oscillation frequency, the amplitude and thus the beam deflection angle decreases. At 2kHz, the max. beam deflection angle is ca.  $50\mu\text{rad}$ .

The same setup as shown in Figure 2 was used for the dynamic measurements, but with the bare mirror replaced by the full FSM including actuators and corresponding electronics. Reducing the camera shutter time of the wavefront sensor to ca.  $40\mu\text{s}$  and increasing the light source power accordingly allows to measure the reflected wavefront and thus the FSM surface shape aberrations while the FSM oscillates, up to an oscillation frequency of 2kHz. The measurement results presented in Table 3 confirm that indeed the actuation of the FSM has no significant impact on the surface shape of the mirror.

Table 3: The surface shape aberrations of the FSM do not show a dependence on the oscillation frequency in the frequency range up to 2kHz.

| Oscillation Frequency / Hz | 10    | 30    | 60    | 100   | 200   | 500   | 1000  | 2000  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| RMS / $\mu\text{m}$        | 0.036 | 0.036 | 0.038 | 0.038 | 0.036 | 0.038 | 0.038 | 0.037 |
| PV / $\mu\text{m}$         | 0.153 | 0.154 | 0.173 | 0.166 | 0.156 | 0.163 | 0.163 | 0.162 |

### 4. MEASUREMENT IN VACUUM CONDITION

In space, the FSM operates under vacuum conditions. The above described studies have been performed under ambient conditions. To analyze the mirror's performance under vacuum conditions, the FSM was installed in a thermal vacuum chamber. The surface shape aberrations of the mirror were measured using the Shack-Hartmann wavefront sensor setup as

shown in Figure 4, before and after evacuation of the chamber at a pressure of less than  $10^{-3}$  mbar. The Shack-Hartmann wavefront sensor was placed outside the vacuum chamber.

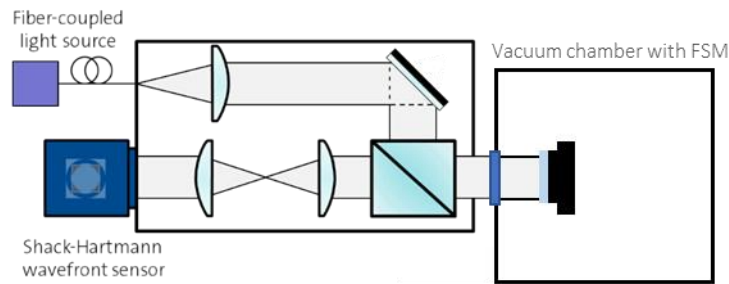


Figure 4: Setup for testing of plano mirrors in reflected light configuration using a Shack-Hartmann wavefront sensor

The measurement was done to compare the change of the mirror surface shape before and after evacuation, i.e. a relative measurement was performed. This change was found to be small and in the same range as the variations observed in the repeatability measurements described in section 2.

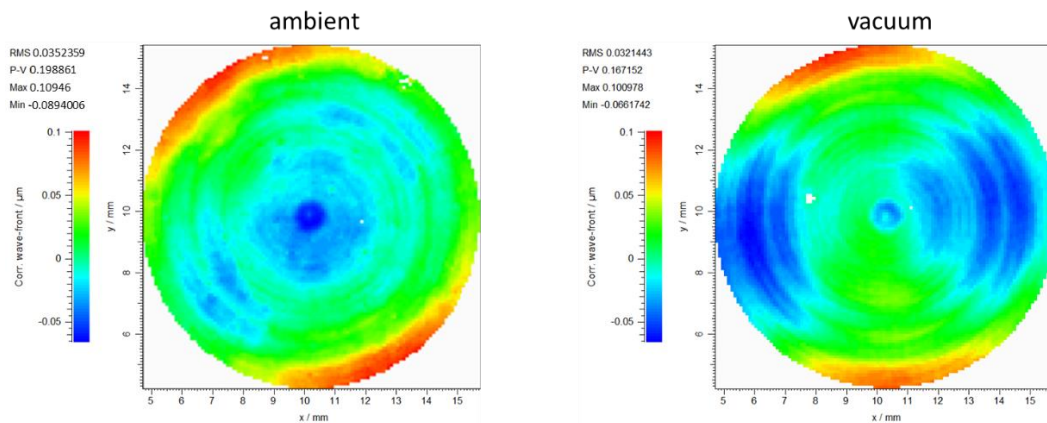


Figure 5: Mirror surface shape in ambient conditions (left) and in vacuum conditions after evacuation of the chamber (right).

## 5. CONCLUSION

Phase shifting deflectometry and Shack-Hartmann wavefront metrology have been proven to be suitable measurement methods characterizing the surface shape aberrations of fast steering mirrors. While PSD offers a higher lateral resolution and is thus capable of detecting smaller features, the WFS retrieves the surface shape information from a single camera frame and thus enables fast dynamic measurements. The measurement precision of the WFS is slightly higher for the three mirrors investigated in the present study. In terms of absolute surface shape aberrations, both measurement approaches agree well within the given precision range.

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